

Regression Documentation: Drug Consumption Dataset.

Daniel Marquino Pérez, 819028
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1. Introduction and Research Question

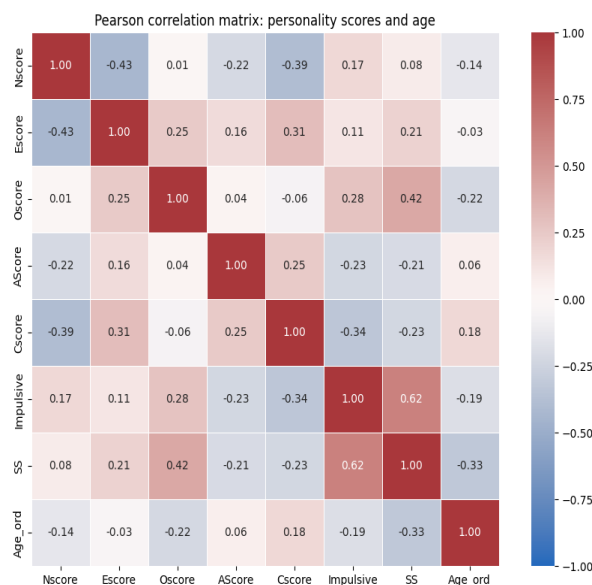
The Drug Consumption dataset contains 1,884 respondents and 31 features: **5 demographic variables**, **7 z-standardized personality scores**, and **19 ordinal drug-use columns** (scale CL0–CL6). The personality scores (*Nscore*, *Escore*, *Oscore*, *AScore*, *CScore*, *Impulsive*, *SS*) are the only continuous features and are therefore the natural candidates for a regression analysis.

Two findings from the exploratory phase motivate the task. First, **Sensation Seeking (SS) declines monotonically with age**. Second, **the personality scores are mutually correlated**, with *Impulsivity* and *SS* visibly moving together. Because *SS* was the most informative trait in previous phases, the strongest correlate of substance use in the EDA and the top-ranked feature in the classification, the regression phase aims to answer **what explains SS itself?** For this reason, **SS is chosen as the dependent variable**.

Three models are compared, each answering a distinct question: a **simple linear regression** ($SS \sim \text{Impulsive}$) as an interpretable baseline, a **multiple linear regression** ($SS \sim \text{six remaining personality scores} + \text{Age}$) to measure each trait's contribution and a **polynomial (generalised linear) regression** ($SS \sim \text{Age}$) to test whether the age-related decline is linear or curved.

2. Continuous Features and Correlation Analysis

Pairwise Pearson correlations among the seven personality scores and ordinal Age were computed as a screening step (Figure 1). Age was encoded as equally spaced ranks ($18-24 = 0, \dots, 65+ = 5$) and treated as quasi-interval; its correlations are approximate but adequate for feature selection.



Three relationships involving SS stand out: *Impulsive-SS* ($r = 0.623$), the strongest pair in the matrix, *Oscore-SS* ($r = 0.422$) and *SS-Age* ($r = -0.327$), the decline already observed graphically in the EDA. *SS* participates in the three strongest relationships of interest and acts as the center of the correlation structure, which **confirms it as the appropriate dependent variable**.

Excluding spurious correlation (a correlation is spurious when a third variable drives both).

The most plausible confounder here is **Age**, since both *Impulsivity* and *SS* decline with age.

Figure 1: Heatmap

The partial correlation of *Impulsive* and *SS* controlling for *Age* is 0.606 , almost identical to the raw 0.623 , so the relationship is **not** based on age and a spurious correlation through age can be excluded.

A conceptual caveat remains: both scales originate from the impulsivity domain of psychology, so part of the association is overlap between related constructs rather than an independent mechanism.

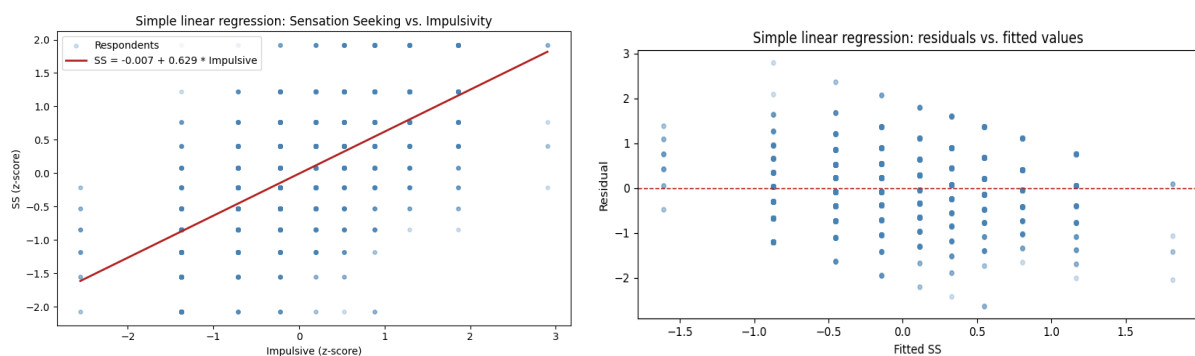
3. Methodology

All three models were evaluated with **5-fold cross-validation** so that every model is scored on data it was not fitted on, plain KFold is used instead of the stratified variant used previously

because the target is continuous. Out-of-fold predictions produced three complementary criteria: **R²** (*share of explained variance*), **RMSE** (*typical error, penalizing large deviations*) and **MAE** (*typical error, robust to outliers*). *Country*, *Ethnicity* and *Gender* were deliberately excluded: the research question focuses on the personality profile, and *Country* was shown to act mainly as a sampling-bias shortcut.

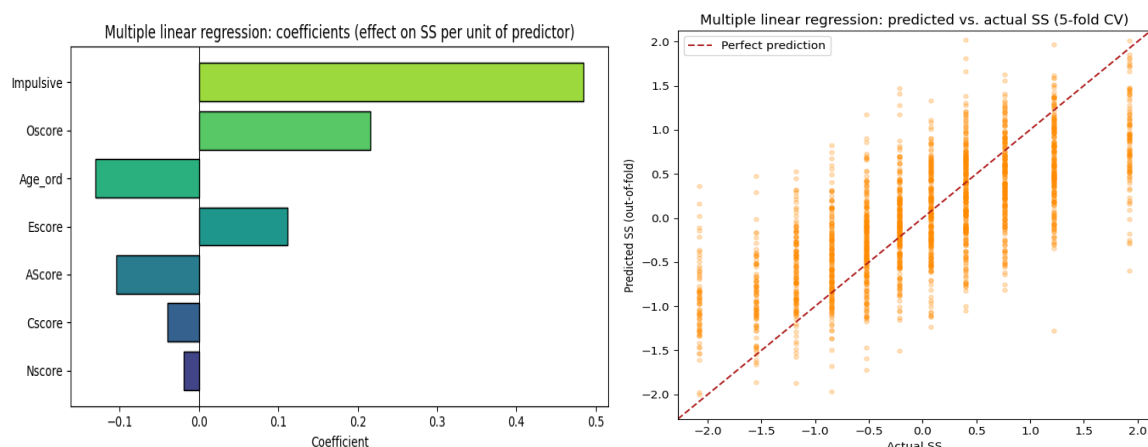
4. Results

Model 1: Simple linear regression ($SS \sim Impulsive$). The fitted equation is $SS = -0.007 + 0.629 \times Impulsive$: a one-standard-deviation rise in *Impulsivity* is associated with a 0.63 SD rise in *SS*. The model explains $R^2 = 0.388$ on the full data, with a near-identical cross-validated $R^2 = 0.385 \pm 0.029$, a one-parameter model can't overfit 1,884 observations. The vertical banding shown in Fig. 2 is caused by the data organization, as it takes only a few discrete values. The residuals-versus-fitted plot (Fig. 3) is centered on zero with no funnel shape or curvature, confirming the linear form is adequate.



Figures 2 & 3: *SS vs. Impulsivity* & *residuals vs. fitted*.

Model 2: Multiple linear regression (7 predictors). Pooling all traits and Age raises the explained variance to $R^2 = 0.505$ (*cross-validated* 0.500 ± 0.032), with RMSE = 0.680. The adjusted coefficients (Fig. 4) are informative: ***Impulsive* (+0.485) remains dominant but decreases** from 0.629, because part of its effect was shared with *Openness* and *Age*. *Oscore* (+0.217) and *Escore* (+0.111) add genuine independent signal, *Age* (-0.131 per bracket) confirms the EDA decline, *Nscore* (-0.019) and *Cscore* (-0.040) contribute almost nothing once the other traits are controlled. The predicted-versus-actual plot (Fig. 5) shows the characteristic signature of $R^2 \approx 0.5$: clear alignment along the diagonal, but predictions compressed toward the mean, so extreme *SS* values are rarely reproduced.



Figures 4 & 5: *effect on SS* & *expected vs. actual*.

Model 3: Polynomial regression ($SS \sim Age$). Cross-validated R^2 is essentially flat across degrees 1-4 (0.098 - 0.105), every gain beyond degree 1 is far smaller than one standard deviation of the CV scores. By the parsimony principle, degree 2 is retained only to demonstrate that curvature is negligible. The linear and quadratic fits are visually indistinguishable, and both track the age-group means closely (Fig. 6), with only a hint of flattening in the two oldest groups:

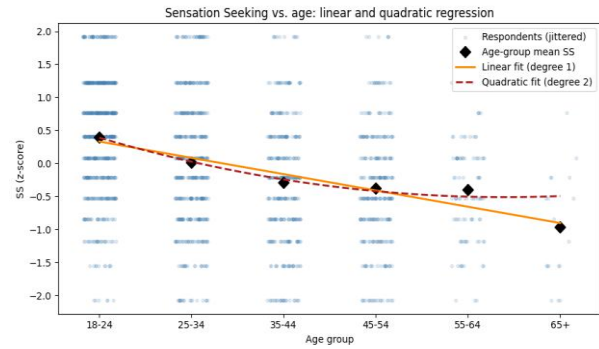


Figure 6: linear vs. quadratic regression.

the decline of SS with age is **essentially linear**, resolving the functional-form question left open by the EDA. However, *Age* alone explains only $\approx 10\%$ of the SS variance, the wide within-group scatter shows the age effect is systematic at the group level but weak at the individual level.

5. Model Comparison

Model	CV R^2	CV RMSE	CV MAE
1. Simple linear ($SS \sim Impulsive$)	0.385 ± 0.029	0.754	0.597
2. Multiple linear (7 predictors)	0.500 ± 0.032	0.680	0.537
3. Polynomial ($SS \sim Age$, degree 2)	0.103 ± 0.044	0.909	0.723

6. Predictions for New Data

The two hypothetical respondents from the previous phase were reused to keep the test comparable. In Phase 2 their SS values were assumed as part of the profile design ($+1.5$ for the *high-risk respondent*, -1.5 for the *low-risk respondent*). Here the models predict SS from the remaining traits, so the assumed values serve only as a plausibility reference.

Respondent	Model 1 (simple)	Model 2 (multiple)	Assumed
A (young, high Impulsive/Openness)	+0.748	+1.139	+1.5
B (older, low Impulsive/Openness)	-0.636	-1.129	-1.5

The direction and order are correct: both models give positive SS for the high-risk profile and negative for the low-risk profile, also all values lie inside the observed SS range (-2.08 to $+1.92$). The attenuation toward the mean ($\approx +1.14$ / -1.13 against the assumed ± 1.5) is exactly what a model with $R^2 = 0.5$ should produce, it predicts the *expected* SS given the profile, treating extreme profiles conservatively. The multiple model also separates the two respondents more strongly than the simple model (2.27 vs. 1.38 z-score units apart), because it additionally exploits *Openness* and *Age*.